Project 98: University Academic Progression Audit

Python Mini Project - College Assignment

Aim

To check semester marks, subject pass conditions, attendance and prerequisite completion, determine progression eligibility, and identify outstanding requirements.

Algorithm

1. Start the program.
2. Store student names, semester marks, attendance, and prerequisite records.
3. Set the minimum pass mark and attendance percentage.
4. For each student, calculate total and average marks.
5. Check failed subjects, attendance shortage, and incomplete prerequisites.
6. Mark the student eligible only when all conditions are satisfied.
7. Display each student's status and outstanding requirements.
8. Count eligible and ineligible students; print the final summary.
9. Stop.

Eligibility Rules

Requirement	Condition
Subject marks	At least 40 in every subject
Attendance	At least 75%
Prerequisites	All completed
Progression	All three conditions satisfied

Complete Python Program

```
# Project 98: University Academic Progression Audit

print("====================================")
print(" UNIVERSITY ACADEMIC PROGRESSION AUDIT")
print("====================================")

# Student names
students = ["Ravi", "Kiran", "Anil", "Sita"]

# Semester marks (3 subjects)
marks = [
    [75, 80, 65],
    [35, 60, 55],
    [70, 68, 72],
    [45, 50, 38]
]

# Attendance percentages
attendance = [85, 78, 65, 90]

# Prerequisite completion
```

```
# 1 = Completed, 0 = Not Completed
prerequisites = [1, 1, 0, 1]

# Academic requirements
pass_mark = 40
minimum_attendance = 75

eligible_count = 0
not_eligible_count = 0

print("\nACADEMIC PROGRESSION REPORT")

for i in range(len(students)):
    print("\n----------------------------------")
    print("Student:", students[i])

    total = 0
    failed_subjects = 0

    # Check subject marks
    for j in range(len(marks[i])):
        print("Subject", j + 1, "Marks:", marks[i][j])
        total += marks[i][j]
        if marks[i][j] < pass_mark:
            failed_subjects += 1

    average = total / len(marks[i])
    print("Total Marks:", total)
    print("Average Marks:", round(average, 2))
    print("Attendance:", attendance[i], "%")

    # Check eligibility
    eligible = True

    if failed_subjects > 0:
        eligible = False
        print("Failed Subjects:", failed_subjects)

    if attendance[i] < minimum_attendance:
        eligible = False
        print("Attendance Shortage")

    if prerequisites[i] == 0:
        eligible = False
        print("Prerequisite Not Completed")

    if eligible:
        print("Progression Status: ELIGIBLE")
        eligible_count += 1
    else:
        print("Progression Status: NOT ELIGIBLE")
        not_eligible_count += 1

print("\n======================================")
print("       FINAL PROGRESSION SUMMARY")
print("======================================")
print("Total Students:", len(students))
print("Eligible Students:", eligible_count)
print("Not Eligible Students:", not_eligible_count)
print("\n======================================")
print("          AUDIT COMPLETED")
print("======================================")
```

Sample Output (from running the program)

```
======================================
 UNIVERSITY ACADEMIC PROGRESSION AUDIT
======================================

ACADEMIC PROGRESSION REPORT
```

```
---------------------------------
Student: Ravi
Subject 1 Marks: 75
Subject 2 Marks: 80
Subject 3 Marks: 65
Total Marks: 220
Average Marks: 73.33
Attendance: 85 %
Progression Status: ELIGIBLE

---------------------------------
Student: Kiran
Subject 1 Marks: 35
Subject 2 Marks: 60
Subject 3 Marks: 55
Total Marks: 150
Average Marks: 50.0
Attendance: 78 %
Failed Subjects: 1
Progression Status: NOT ELIGIBLE

---------------------------------
Student: Anil
Subject 1 Marks: 70
Subject 2 Marks: 68
Subject 3 Marks: 72
Total Marks: 210
Average Marks: 70.0
Attendance: 65 %
Attendance Shortage
Prerequisite Not Completed
Progression Status: NOT ELIGIBLE

---------------------------------
Student: Sita
Subject 1 Marks: 45
Subject 2 Marks: 50
Subject 3 Marks: 38
Total Marks: 133
Average Marks: 44.33
Attendance: 90 %
Failed Subjects: 1
Progression Status: NOT ELIGIBLE

====================================
       FINAL PROGRESSION SUMMARY
====================================
Total Students: 4
Eligible Students: 1
Not Eligible Students: 3

====================================
          AUDIT COMPLETED
====================================
```

Flowchart (text representation)

```
START
  |
Store marks, attendance, prerequisites
  |
Calculate total and average
  |
All subjects passed? -- No --> NOT ELIGIBLE
  | Yes
Attendance >= 75% and prerequisites completed?
  | Yes                    | No
ELIGIBLE               NOT ELIGIBLE
  |                        |
  +------> Display summary <-+
```

Python Concepts Used

Variables, integers, floats, strings, booleans, arithmetic and comparison operators, if/else statements, for loops, lists and nested lists.

Result

Total students: **4**. Eligible: **1 (Ravi)**. Not eligible: **3**. Kiran and Sita each failed one subject; Anil has an attendance shortage and incomplete prerequisites. These eligibility rules are illustrative and may differ from actual university policies.